My K Library: Programmer's UI Coding Manual (V15 - Redesign)

This manual covers the UI Components Redesign for V15, including resolution scaling, modern components, and layout persistence.

1. Resolution Manager

The UI relies on `resolutionMode` stored in the `Settings` table (Small, Medium, Large). When initialized, `ApplyResolutionSize()` configures the main window dimensions, and scales the base font up/down dynamically. Buttons generated via `CreateAutoBtn` pull height logic and font scaling from this specific mode.

2. Dynamic FlowLayout Buttons

All toolbars (Header, Welcome Actions, Library Actions, Scan Row) have been upgraded to `FlowLayoutPanel` structures. Buttons configure `AutoSize = True` and specific `Padding` metrics so text never gets clipped regardless of language or screen width.

3. Z-Index and Docking Flow

The DataGridView bugs encountered during refactoring were deeply tied to WinForms internal Z-Index docking algorithms.

- `SendToBack()` forces top-docked panels (like nav row) to secure vertical space first.
- `BringToFront()` forces the core `pnlGridArea` and `dgvBooks` to evaluate **last**, allowing them to perfectly consume the remaining client screen space without hiding behind standard panels.

4. Visual Studio WinForms Designer

We introduced MSBuild `.vbproj` integration. For Visual Studio to load the designer accurately without splitting the `Program.vb` file into two, the `Module Program` has been relocated to the **bottom** of the file, allowing `MainForm` to act as the primary graphical type for designer tools.

5. UI Component Coding System Index

Each GUI component initialized in `Program.vb` is tagged with a universal `CODE-XXXXX` marker to make component modifications and stylistic overrides globally searchable across visual and logical modules.

Code Tag	Component / Element Description
CODE-00001	Application Header & Logo Block
CODE-00002	Main Navigation Dropdown (lblNav, cmbNavigation)
CODE-00003	Global Application Settings & Export Buttons Container
CODE-00004	Grid Style Config Toolbar Action
CODE-00005	Harvest Covers Action Button
CODE-00006	Export CSV Database Action
CODE-00007	Browser Web Server UI Launcher
CODE-00008	Scale Fonts Engine Actions
CODE-00009	User Settings Configuration Launcher
CODE-00010	Statistical Dashboard Title & Base Metrics Card
CODE-00011	Realtime Additions Scrolling Ticker
CODE-00012	Welcome Screen Action Panel (Library Select Dropdown)
CODE-00013	Library Management Buttons Group (Add, Rename, Delete, Merge)
CODE-00014	Main View Area Layout Container (pnlLibrary)
CODE-00015	Write Lock Authentication Mechanism & Current Book Selector
CODE-00016	Direct Record Action Row (Add, Edit, Delete, Search Books)
CODE-00017	ISBN Hardware Scanner Implementation Input Row
CODE-00018	Gemini AI Smart Metadata Lookup Trigger
CODE-00019	Core Database Record DataGridView Component Layer
CODE-00020	Image Previews Dual-Sidebar Block (Cover / Back Graphics)